

Lab 2 Solutions

Statistics for Laboratory Scientists II

1. (a) This concerned data on survival according to cancer type.

Preparations:

```
# data
cancer <- read.csv("cancer.csv")

# Make sure that "type" is a factor
sapply(cancer, is.factor)
```

This is a one-way ANOVA situation. We do the ANOVA and then study the diagnostic plots.

```
# Apply one-way ANOVA
aov.out <- aov(response ~ type, data=cancer)

# Look at diagnostic plots.
plot(aov.out)
```

The variance is clearly not constant, and the residuals are not normally distributed. We'll try taking logs.

```
# take logs
cancer$response <- log10(cancer$response)

# re-do the ANOVA
aov.out <- aov(response ~ type, data=cancer)

# re-do the diagnostic plots
plot(aov.out)
```

Now everything looks fine. We'll go ahead and calculate the ANOVA table.

```
summary(aov.out)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
type	4	4.62	1.15	4.29	0.0041
Residuals	59	15.89	0.27		

We get a P-value of 0.004, indicating strong evidence for differences among the cancer types.

- (b) This concerned data on the effect of oxygen level on fermentation end productions.

Preparations:

```
ferm <- read.csv("fermentation.csv")

# check that things are factors that should be
sapply(ferm, is.factor)

# make "oxygen" a factor
ferm$oxygen <- factor(ferm$oxygen)
```

This is a two-way ANOVA problem: ethanol response as a function of oxygen level and sugar type. We do the two-way ANOVA and then look at the diagnostic plots.

```
aov.out <- aov(ethanol ~ oxygen * sugar, data=ferm)

# Do the diagnostic plots
plot(aov.out)
```

Again, we see non-constant variance. This time, let's take square roots.

```
ferm$ethanol <- sqrt(ferm$ethanol)

# Re-do the two-way ANOVA
aov.out <- aov(ethanol ~ oxygen * sugar, data=ferm)

# Re-do the diagnostic plots
plot(aov.out)
```

Now everything looks okay, so we'll calculate the ANOVA table.

```
summary(aov.out)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
oxygen	3	0.19	0.064	3.01	0.094
sugar	1	0.40	0.40	18.62	0.0026
oxygen:sugar	3	0.0014	0.00047	0.022	0.995
Residuals	8	0.17	0.021		

Our conclusions: there is a clear sugar effect; there is some evidence for an oxygen effect, but it is not significant; and there doesn't appear to be an interaction.

2. The final analysis: the effect of caffeine on performance.

Preparations:

```
caffeine <- read.csv("caffeine.csv")

# check whether things are factors that should be
supply(caffeine, is.factor)

# make the "dose" column a factor
caffeine$dose <- factor(caffeine$dose)
```

(a) We first do a one-way ANOVA.

```
aov.out <- aov(response ~ dose, data=caffeine)
plot(aov.out)
summary(aov.out)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
dose	2	61.4	30.7	6.18	0.0062
Residuals	27	134.1	5.0		

We conclude that there really are differences between the doses.

- (b) We use the function `ci.bonf` defined in the notes for lecture 11 (`comp11.R`) to calculate the Bonferroi-corrected confidence intervals.

```
ci.bonf(caffeine$response, caffeine$dose)
```

	est	lower	upper
0 - 100	-1.6	-3.8	0.6
0 - 200	-3.5	-5.7	-1.3
100 - 200	-1.9	-4.1	0.3

We conclude that the response to the 200mg dose is larger than that of the 0mg dose, but that there are no significant differences between the 0mg and 100mg doses or the 100mg and 200mg doses.

- (c) We use the function `TukeyHSD()` using the output from `aov()` to get Tukey's HSD confidence intervals.

```
tuk <- TukeyHSD(aov.out)
```

	diff	lwr	upr
100-0	1.6	-0.9	4.1
200-0	3.5	1.0	6.0
200-100	1.9	-0.6	4.4

Our conclusions are the same as for part (b).